

# Agenda

## 1. Announcements

- \* Homework will follow the course schedule (even though we're a day ahead)

- \* Module 7 has extra 7.4 examples

- \* If you can, please look at Module 8 before Friday

- \* MLC study skills workshops start this week!

## 2. Remaining Activity/Example from Friday

## 3. Partial Fraction Decomposition - Section 7.4

★ Activity: For each integral, identify the best technique (basic, u-sub, IBP, trig int, trig sub, no tool yet)

$$\int \frac{\ln(x)}{x} dx$$

$= \int \ln(x) \left(\frac{1}{x}\right) dx$   
 $u = \ln x$   
 u-sub

$$\int \tan^3 \theta \sec^4 \theta d\theta$$

$\int \tan^2 \theta (\tan \theta + 1) \sec^2 \theta d\theta$   
 trig sub

$$\int \sin^3 \theta d\theta$$

$(1 - \cos^2 \theta) \sin \theta d\theta$   
 $u = \cos \theta$   
 trig sub

$$\int \frac{1}{\sqrt{x^2 - 6x + 45}} dx$$

trig sub  
complete the square

$$\int \frac{x^2 - 2x + 3}{x^2 - 1} dx$$

X fac decomp  
 (7.4)

$$\int \cos^4 \theta d\theta$$

double angle formula x2  
 trig integral

$$\int_0^1 \frac{t-4}{t^2-5t+6} dt$$

(7.4)

$$\int x^3 \sqrt{9-x^2} dx$$

u-sub

$$\int (\ln(t))^2 dt$$

IBP

$$\int x^2 \sqrt{9-x^2} \cdot x dx$$

$u = 9-x^2$   
 $x^2 = 9-u$   
 $\frac{1}{2} \int (9-u) u^{1/2} du$

$$u = \ln t^2$$

$du = dt$

Example:  $\int \frac{1}{\sqrt{x^2 - 6x + 45}} dx$

$$(x-3)^2 = x^2 - 6x + 9$$

+36 +36

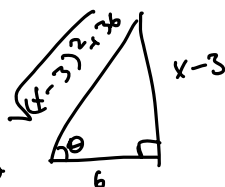
$$(x-3)^2 + 36 = x^2 - 6x + 45$$

$$= \int \frac{1}{\sqrt{(x-3)^2 + 36}} dx$$

$\tan \theta = \frac{x-3}{6}$

$$x-3 = 6 \tan \theta$$

$$dx = 6 \sec^2 \theta d\theta$$



(. . .)

## Integrals using Partial Fraction Decomposition

Partial fraction decomposition moves a rational expression form into several fractions.

$$\frac{-x+23}{x^2-x-6} = \frac{-x+23}{(x+2)(x-3)} = \frac{A}{(x+2)} + \frac{B}{(x-3)} (x^2-x-6)$$

$$-x+23 = A(x-3) + B(x+2)$$

$$\text{if } x=3$$

$$-3+23 = \cancel{A(0)} + B(3+2)$$

$$\frac{4 \cdot 20}{5} = B$$

$$\text{if } x=-2$$

$$2+23 = A(-2-3)$$

$$A = -5$$

$$\frac{-5}{x+2} + \frac{4}{x-3}$$

## Distinct Linear Factors

Let  $a$  be a constant. Then we have

$$\int \frac{dx}{x+a} = \int \frac{du}{u} = \ln|u| + C = \ln|x+a| + C$$

can skip u-sub

Example:  $\int \frac{3}{x^2+x-2} dx = \int \frac{3}{(x+2)(x-1)} dx$

$$= \frac{A}{x+2} + \frac{B}{x-1} (x^2+x-2)$$

$$3 = A(x-1) + B(x+2)$$

choose  $x = 1$

$$3 = \cancel{A(0)} + B(1+2)$$

$$\boxed{B = 1}$$

—  $x = -2$

$$3 = A(-3) + \cancel{B(0)}$$

$$\boxed{A = -1}$$

$$= \int \frac{-1}{x+2} + \frac{1}{x-1}$$

$$= -\ln|x+2| + \ln|x-1| + C$$

Activity:  $\int \frac{dx}{x(x-5)} = \frac{A}{x} + \frac{B}{x-5}$

$$1 = A(x-5) + B(x)$$

$$1 = A(0) + B(5)$$

$$\boxed{B = 1/5}$$

$x = 0$

$$1 = A(-5) + \cancel{B(0)}$$

$$\boxed{A = -1/5}$$

$$= \int \frac{-1/5}{x} + \frac{1/5}{x-5} dx$$

integrate:

$$= \frac{1}{5} \left| \frac{1}{x} + \int \frac{1}{x-5} \right|$$

$$= \frac{1}{5} \ln|x| + \frac{1}{5} \ln|x-5| + C$$

## Repeated Linear Factors

Example:  $\int \frac{x}{(x-4)^2} dx$  ← only find C for this

$$\frac{x}{(x-4)^2} = \frac{A}{(x-4)} + \frac{B}{(x-4)^2}$$

$$x = A(x-4) + B$$

(...)

Example:  $\int \frac{8dx}{x(x-2)^2}$

$$\frac{8}{x(x-2)^2} = \frac{A}{x} + \frac{B}{x-2} + \frac{C}{(x-2)^2}$$

$$8 = A(x-2)^2 + Bx(x-2) + Cx$$

$$\left. \begin{array}{l} \underline{x=2}: \dots C=4 \\ \underline{x=0}: \dots A=2 \end{array} \right\} \underline{x=1}: \left( \begin{array}{l} \text{plus} \\ C=4, A=2 \end{array} \right) \cdot \underline{B=-2}$$

and any third value ...  
can determine

## General Form for Linear Factors

Example:  $\frac{3x^3 - 2x^2 + 4x - 4}{x^3(x+1)(x-5)^2} =$

Degree of Numerator  $\geq$  Degree of Denominator

Step 1: Polynomial Long Division

Step 2: Use Partial Fraction Decomposition on Remainder

Example:  $\int \frac{x^2 + 2x}{x^2 - 1} dx$